into another circuit without changing the frequency.

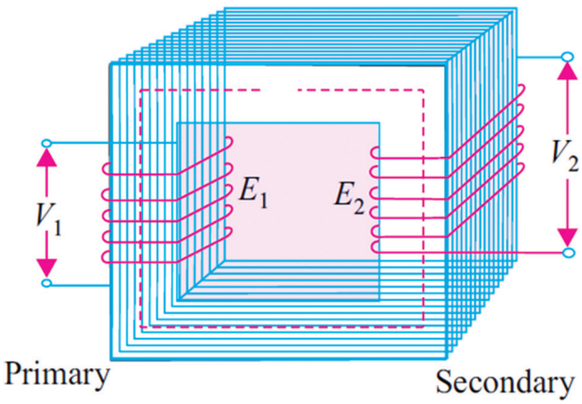


Fig 6.4 *Construction of an Ideal Transformer*

It consists of two inductive coils which are electrically separated but magnetically coupled to a core as shown in Fig. 6.4. It operates on the principle of mutual induction between two (or) more inductively coupled coils. If the coil is connected to a AC source, an alternating flux is setup. Most of the flux is linked with the other coil. This flux is called mutual flux.

As per Faraday's laws of electromagnetic induction, an emf is induced in the second coil. The coil (First coil) which is connected to the A.C supply is called as primary winding. The coil (Second coil) which is connected to the load is called as secondary winding.

6.3.1 STEP UP/ STEP DOWN TRANSFORMER:

Step up Transformer – If the output voltage in secondary winding is higher than the input voltage, it is called step up transformer. (See Figure 6.5)

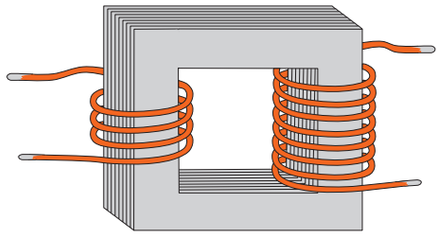


Fig 6.5 *Step-up Transformer*

Step down Transformer – If the output voltage in the secondary winding is lesser than the input voltage, it is called step down transformer as in fig. 6.6.

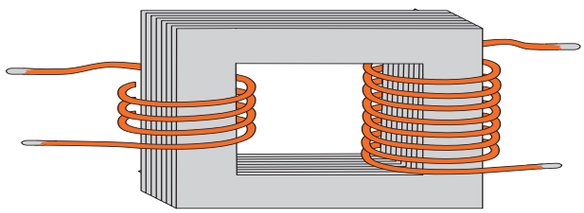


Fig 6.6 *Step-down Transformer*

6.3.2 ADVANTAGES OF TRANSFORMERS

i. The transformer is a static machinery. Hence there is no wear and tear and no friction losses in it.
ii. Maintenance cost is low.
iii. As there is no rotating part in it, extra high voltage can be transferred easily by providing a good insulation to its winding.

6.4 E.M.F EQUATION OF A TRANSFORMER

a) Construction b) Wave Form

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